

Lecture 8:

AI504 - Knowledge Representation

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1 Relative clauses

The logic of $\mathcal{A}(RC)$ of all $_+ +$ relative clauses)

Previously, a **signature** was simply a set of nouns, now our signature consists of a set of nouns, **plus** a set of transitive verbs (a verb must take an object).

- **What are sentences? What are models? What are proofs?**

1.1 Models

A Model (*relative to a particular signature*) consists of a set (*the domain*), and interpretations of both all nouns and all verbs

- $\llbracket _ \rrbracket_{\text{nouns}}$ is a subset of the domain
- $\llbracket _ \rrbracket_{\text{verbs}}$ is a binary relation on the domain

so..

$$\llbracket _ \rrbracket_{\text{nouns}} : P \rightarrow \mathcal{P}(M), \quad \llbracket _ \rrbracket_{\text{verbs}} : R \rightarrow \mathcal{P}(M \times M)$$

Where P is the set of nouns, and R is the set of verbs

so..

$$\mathcal{M} = (P, \llbracket _ \rrbracket_{\text{nouns}} : P \rightarrow \mathcal{P}(M), \quad \llbracket _ \rrbracket_{\text{verbs}} : R \rightarrow \mathcal{P}(M \times M))$$

1.2 Terms

We want to give meaning to relative clauses so we ignore irrelevant words.

~~those who~~ love all ~~who~~ fear all woodchucks
 love all (fear all woodchucks)

A term is either

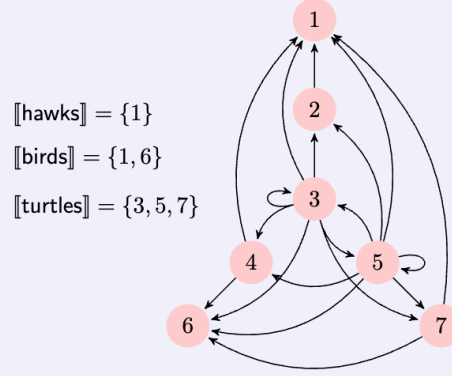
- A single noun, or
- an expression $(r \text{ all } t)$, where r is a verb and t is a term

Terms

- | | | |
|---|----|---|
| | i | |
| 1 | | if t is a single noun, then |
| | | $\llbracket t \rrbracket$ is simply $\llbracket t \rrbracket_{\text{noun}}$, which comes from the data in $\mathcal{M}$ |
| | ii | |
| 2 | | if t is $r \text{ all } t'$, then |
| 3 | | $\llbracket t \rrbracket = \{m \in M : \forall n \in \llbracket t' \rrbracket \quad (m, n) \in \llbracket r \rrbracket\}$ |

1.2.1 Example

Consider the model $\mathcal{M}$ with universe $\{1, \dots, 7\}$, and with interpretations of three nouns, hawks, birds, and turtles, and one verb, see, shown below.



$\llbracket \text{hawks} \rrbracket = \{1\}$
 $\llbracket \text{birds} \rrbracket = \{1, 6\}$
 $\llbracket \text{turtles} \rrbracket = \{3, 5, 7\}$

In this model,

$\llbracket \text{see all hawks} \rrbracket = \{2, 3, 4, 5, 7\}$	$\llbracket \text{see all turtles} \rrbracket = \{3, 5\}$
$\llbracket \text{see all birds} \rrbracket = \{3, 4, 5, 7\}$	$\llbracket \text{see all (see all hawks)} \rrbracket = \{3, 5\}$

Here are some examples of sentences false and true in the model:

$\mathcal{M} \not\models \text{All (see all hawks) (see all birds)}$
 $\mathcal{M} \models \text{All (see all turtles) (see all (see all hawks))}$
 $\mathcal{M} \models \text{All (see all (see all hawks)) (see all turtles)}$

Figure 1: Example 2.4 from: Logic From Language by Lawrence S. Moss

$$\llbracket \text{see all hawks} \rrbracket = \{m \in M : \forall n \in \{1, 2, 3, 4, 5, 6\} \quad (m, n) \in \{1\}\} = \{2, 3, 4, 5, 7\}$$

$$\llbracket \text{see all birds} \rrbracket = \{m \in M : \forall n \in \{1, 2, 3, 4, 5, 6\} \quad (m, n) \in \{1, 6\}\} = \{3, 4, 5, 7\}$$

$$\llbracket \text{see all turtles} \rrbracket = \{m \in M : \forall n \in \{1, 2, 3, 4, 5, 6\} \quad (m, n) \in \{3, 5, 7\}\} = \{3, 5\}$$

$$\llbracket \text{see all (see all hawks)} \rrbracket = \{m \in M : \forall n \in \{2, 3, 4, 5, 7\} \quad (m, n) \in \{1\}\} = \{3, 5\}$$

1.3 Sentences

A **sentence** is an expression

All term term

1.4 Fundamental Lemma

Work within a fixed model.

For any terms s and t and any ver r . if $\llbracket s \rrbracket \subset \llbracket t \rrbracket$

$$\llbracket r \text{ all } t \rrbracket \subset \llbracket r \text{ all } s \rrbracket$$

1.4.1 Proof

Suppose that m is an arbitrary element of M .

Assume that $m \in \llbracket r \text{ all } t \rrbracket$ Goal: $m \in \llbracket r \text{ all } s \rrbracket$

This means that $\forall n \in \llbracket t \rrbracket \quad (m, n) \in \llbracket r \rrbracket$ Goal: $\forall n \in \llbracket s \rrbracket \quad (m, n) \in \llbracket s \rrbracket$

Now let n be an arbitrary element of $\llbracket s \rrbracket$ Goal: $(m, n) \in \llbracket s \rrbracket$

Since $\llbracket s \rrbracket \subset \llbracket t \rrbracket$, $n \in \llbracket t \rrbracket$

Now because of $\forall n \in \llbracket s \rrbracket \quad (m, n) \in \llbracket s \rrbracket$ and $n \in \llbracket t \rrbracket$

We know that $(m, n) \in \llbracket r \rrbracket$

□